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POWER DISTRIBUTION UNIT

Abstract

A power distribution unit includes a plurality of outlets and a controller. The controller is configured to poll the plurality of outlets to determine a polled electrical load of each outlet, add one or more outlets of the plurality of outlets to a queue based on a comparison of the polled electrical loads and a user set threshold, connect the queued outlets to a power input, monitor a total electrical load of the power distribution unit, and disconnect one or more outlets of the plurality of outlets from the power input in response to determining that the total electrical load exceeds the user set threshold.

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Background/Summary

FIELD

[0001] The present disclosure relates to power distribution devices.

SUMMARY

[0002] Embodiments described herein relate to power distribution devices and, more particularly, to power distribution devices for intelligently managing the charging of high-demand devices.

[0003] Charging multiple high-demand devices (such as power tool battery packs) simultaneously on a single circuit can result in a current or power draw that exceeds the circuit's rated capacity. This may cause the circuit breaker to trip, which may result in an interruption in the charging cycle until the circuit breaker can be reset. When the charging cycle is interrupted by a tripped circuit breaker, the charging devices may not receive a full charge, which may lead to reduced device availability and/or operational inefficiency. The interruptions associated with tripped circuit breakers can halt work, or, in the scenario where devices are charged overnight, result in devices being unavailable for use the next morning. Furthermore, frequent trips can also lead to premature wear on the circuit breaker.

[0004] Systems, apparatuses, methods, and techniques described herein effectively solve these and other technical problems by intelligently managing the current draw of multiple devices that may be plugged into a power distribution unit, sequencing the current drawn by each device based on the available current/power capacity, and ensuring that the circuit is not overloaded. When used to charge multiple-high demand devices, power distribution units implemented according to the present disclosure can automate the charging process and ensure that the devices plugged in to the power distribution units are effectively charged over time, eliminating the need for manual intervention and ensuring that the devices are fully charged and ready for use.

[0005] Power distribution units described herein include a plurality of outlets and a controller. The controller is configured to poll the plurality of outlets to determine a polled electrical load of each outlet, add one or more outlets of the plurality of outlets to a queue based on a comparison of the polled electrical loads and a user set threshold, connect the queued outlets to a power input, monitor a total electrical load of the power distribution unit, and disconnect one or more outlets of the plurality of outlets from the power input in response to determining that the total electrical load exceeds the user set threshold.

[0006] In some aspects, to poll the plurality of outlets, the controller is configured to connect a first pair of outlets of the plurality of outlets to the power input, measure an electrical load of each outlet of the first pair of outlets, and disconnect the first pair of outlets of the plurality of outlets from the power input.

[0007] In some aspects, each outlet of the plurality of outlets is assigned a priority. The controller, to poll the plurality of outlets, is configured to connect a second pair of outlets of the plurality of outlets to the power input, the second pair of outlets having a lower priority than the first pair of outlets, measure an electrical load of each outlet of the second pair of outlets, and disconnect the second pair of outlets of the plurality of outlets from the power input.

[0008] In some aspects, the controller, to add one or more outlets of the plurality of outlets to the queue, is configured to select a first outlet of the plurality of outlets, determine whether connecting the first outlet to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold, and add the first outlet to the queue in response to determining that connecting the first outlet to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold.

[0009] In some aspects, each outlet of the plurality of outlets is assigned a priority. The controller, to add one or more outlets of the plurality of outlets to the queue, is configured to select a second outlet of the plurality of outlets, the second outlet having a lower priority than the first outlet, determine whether connecting the second outlet and any queued outlets to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold, and add the second outlet to the queue in response to determining that connecting the second outlet and any queued outlets to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold.

[0010] In some aspects, the controller is configured to disconnect outlets from the power input in reverse priority order in response to determining that the total electrical load exceeds the user set threshold.

[0011] In some aspects, the controller is further configured to determine whether any additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold in response to determining that the total electrical load does not exceed the user set threshold, initiate a timer in response to determining that additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold, and connect the additional outlets to the power input in response to the timer elapsing.

[0012] In some aspects, the power distribution unit includes a plurality of switches, each switch of the plurality of switches being associated with a respective outlet of the plurality of outlets, a plurality of outlet sensors, each outlet sensor of the plurality of sensors being associated with a respective outlet of the plurality of outlets and configured to sense an electrical load of the respective outlet, and a main sensor configured to sense the total electrical load of the power distribution unit at the power input. The controller is configured to actuate each switch of the plurality of switches to connect or disconnect the respective outlet, determine an electrical load of each outlet by monitoring a respective outlet sensor, and determine the total electrical load of the power distribution unit by monitoring the main sensor.

[0013] In some aspects, the polled electrical loads and the total electrical load are current values.

[0014] In some aspects, the polled electrical loads and the total electrical load are power values.

[0015] Methods described herein for controlling a power distribution unit include polling a plurality of outlets of the power distribution unit to determine a polled electrical load of each outlet, adding one or more outlets of the plurality of outlets to a queue based on a comparison of the polled electrical loads and a user set threshold, connecting the queued outlets to a power input, monitoring a total electrical load of the power distribution unit, and disconnecting one or more outlets of the plurality of outlets from the power input in response to determining that the total electrical load exceeds the user set threshold.

[0016] In some aspects, polling the plurality of outlets includes connecting a first pair of outlets of the plurality of outlets to the power input, measuring an electrical load of each outlet of the first pair of outlets, and disconnecting the first pair of outlets of the plurality of outlets from the power input.

[0017] In some aspects, each outlet of the plurality of outlets is assigned a priority. Polling the plurality of outlets includes connecting a second pair of outlets of the plurality of outlets to the power input, the second pair of outlets having a lower priority than the first pair of outlets, measuring an electrical load of each outlet of the second pair of outlets, and disconnecting the second pair of outlets of the plurality of outlets from the power input.

[0018] In some aspects, adding one or more outlets of the plurality of outlets to the queue includes selecting a first outlet of the plurality of outlets, determining whether connecting the first outlet to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold, and adding the first outlet to the queue in response to determining that connecting the first outlet to the power input will not cause the total electrical load of the power distribution unit to

exceed the user set threshold.

[0019] In some aspects, each outlet of the plurality of outlets is assigned a priority. Adding one or more outlets of the plurality of outlets to the queue comprises selecting a second outlet of the plurality of outlets, the second outlet having a lower priority than the first outlet, determining whether connecting the second outlet and any queued outlets to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold, and adding the second outlet to the queue in response to determining that connecting the second outlet and any queued outlets to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold.

[0020] In some aspects, the method includes disconnecting outlets from the power input in reverse priority order in response to determining that the total electrical load exceeds the user set threshold. In some aspects, the method includes determining whether any additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold in response to determining that the total electrical load does not exceed the user set threshold, initiating a timer in response to determining that additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold, and connecting the additional outlets to the power input in response to the timer clapsing.

[0021] In some aspects, the power distribution unit includes a plurality of switches, each switch of the plurality of switches being associated with a respective outlet of the plurality of outlets, a plurality of outlet sensors, each outlet sensor of the plurality of sensors being associated with a respective outlet of the plurality of outlets and configured to sense an electrical load of the respective outlet, and a main sensor configured to sense the total electrical load of the power distribution unit at the power input. The method further includes actuating each switch of the plurality of switches to connect or disconnect the respective outlet, determining an electrical load of each outlet by monitoring a respective outlet sensor, and determining the total electrical load of the power distribution unit by monitoring the main sensor.

[0022] In some aspects, the polled electrical loads and the total electrical load are current values.

[0023] In some aspects, the polled electrical loads and the total electrical load are power values.

[0024] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in application to the details of the configurations and arrangements of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0025] Unless the context of their usage unambiguously indicates otherwise, the articles “a,” “an,” and “the” should not be interpreted as meaning “one” or “only one.” Rather these articles should be interpreted as meaning “at least one” or “one or more.” Likewise, when the terms “the” or “said” are used to refer to a noun previously introduced by the indefinite article “a” or “an,” “the” and “said” mean “at least one” or “one or more” unless the usage unambiguously indicates otherwise.

[0026] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as

a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software-based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components. [0027] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%) of an indicated value.

[0028] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

[0029] Accordingly, in the claims, if an apparatus, method, or system is claimed, for example, as including a controller, control unit, electronic processor, computing device, logic element, module, memory module, communication channel or network, or other element configured in a certain manner, for example, to perform multiple functions, the claim or claim element should be interpreted as meaning one or more of such elements where any one of the one or more elements is configured as claimed, for example, to make any one or more of the recited multiple functions, such that the one or more elements, as a set, perform the multiple functions collectively.

[0030] Other examples, embodiments, features, and aspects will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 illustrates a power distribution unit (“PDU”), according to some embodiments.

[0032] FIG. 2 illustrates a control system for the PDU of FIG. 1, according to some embodiments.

[0033] FIG. 3 illustrates an example process for controlling operation of the PDU of FIG. 1, according to some embodiments.

[0034] FIG. 4 illustrates an example process for polling outlets of the PDU of FIG. 1, according to some embodiments.

[0035] FIG. 5 illustrates an example process for turning on outlets of the PDU of FIG. 1, according

to some embodiments.

[0036] FIG. 6 illustrates an example process for turning on outlets of the PDU of FIG. 1, according to some embodiments.

[0037] FIG. 7 illustrates an example process for turning on outlets of the PDU of FIG. 1, according to some embodiments.

[0038] FIG. 8 illustrates an example process for turning on outlets of the PDU of FIG. 1, according to some embodiments.

[0039] FIG. 9 illustrates an example process for controlling operation of the PDU of FIG. 1, according to some embodiments.

[0040] FIG. 10 illustrates an example process for turning on outlets of the PDU of FIG. 1, according to some embodiments.

[0041] FIGS. 11-13 illustrate an example process for controlling operation of the PDU of FIG. 1, according to some embodiments.

[0042] FIG. 14 illustrates an example process for controlling operation of the PDU of FIG. 1, according to some embodiments.

[0043] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0044] FIG. 1 illustrates a power distribution unit (“PDU”) 100. The PDU 100 includes a body or a housing 105 and an alternating current (“AC”) power input cord 110. The PDU 100 includes a plurality of outlets 115. In the illustrated example, the PDU 100 includes twelve outlets 115-1-115-12, but in other examples, the PDU may include any number of outlets 115. Each outlet 115 may have a priority. For example, the outlet 115-1 may have a highest priority, the outlet 115-2 may have a next highest priority, etc. Each outlet 115 may include one or more indicators 120 associated with the outlet 115. The indicators 120 can provide an indication of, for example, whether the outlet 115 is energized (e.g., is receiving power), a fault condition associated with the outlet 115 is present, etc. The PDU 100 also includes an ON/OFF button 125 for turning the PDU 100 ON and OFF, a display 130, and a user input 135. The display 130 may be configured to display, for example, a user set power threshold for the PDU 100 (e.g., 1000 W) or a user set amperage threshold (e.g., 15A). The user input 135 can be used to increase or decrease the user set power threshold or the user set amperage threshold.

[0045] FIG. 2 illustrates a control system for the PDU 100. The control system includes a controller 200. The controller 200 is electrically and/or communicatively connected to a variety of modules or components of the PDU 100. For example, the illustrated controller 200 is electrically connected to a power input module 205 (e.g., an AC power input module), one or more indicators 210, a display 215 (e.g., display 130), one or more user inputs 220 (e.g., user inputs 135), a first current sensor 225, and a plurality of secondary current sensors 230. Connected between each outlet 115 and each current sensor 230 are a plurality of switches 235 (e.g., relays, semiconductor switches, etc.) for controlling whether power is provided to the outlets 115. In some embodiments, the PDU 100 also includes one or more vents, one or more fans, and one or more temperature sensors for controlling the temperature of the PDU 100. For example, a temperature sensor can be used to measure an internal temperature of the PDU 100. If the temperature is too high or reaches one or more threshold temperature values, the controller 200 can control one or more fans to try to reduce the temperature of the PDU 100.

[0046] The controller 200 includes combinations of hardware and software that are operable to, among other things, control the operation of the PDU 100, monitor the operation of the PDU 100, activate the one or more indicators 210 (e.g., an LED), etc. The current sensor 225 is configured to, for example, sense a total current being drawn by the PDU 100.

[0047] The controller 200 includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller 200

and/or the PDU **100**. For example, the controller **200** includes, among other things, a processing unit **240** (e.g., a microprocessor, a microcontroller, an electronic controller, and electronic processor, or another suitable programmable device), a memory **245**, input units **250**, and output units **255**. The processing unit **240** includes, among other things, a control unit **260**, an arithmetic logic unit (“ALU”) **265**, and a plurality of registers **270** (shown as a group of registers in FIG. 2) and is implemented using a known computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit **240**, the memory **245**, the input units **250**, and the output units **255**, as well as the various modules or circuits connected to the controller **200** are connected by one or more control and/or data buses (e.g., common bus **275**). The control and/or data buses are shown generally in FIG. 2 for illustrative purposes. The use of one or more control and/or data buses for the interconnection between and communication among the various modules, circuits, and components would be known to a person skilled in the art in view of the invention described herein.

[0048] The memory **245** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit **240** is connected to the memory **245** and executes software instructions that are capable of being stored in a RAM of the memory **245** (e.g., during execution), a ROM of the memory **245** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the PDU **100** can be stored in the memory **245** of the controller **200**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **200** is configured to retrieve from the memory **245** and execute, among other things, instructions related to the control processes and methods described herein. In other constructions, the controller **200** includes additional, fewer, or different components.

[0049] The power input module **205** includes combinations of active and passive components to regulate or control the power received from the battery pack prior to power being provided to the controller **200**.

[0050] The indicators **210** include, for example, one or more light-emitting diodes (“LEDs”). The indicators **210** can be configured to display conditions of, or information associated with, the PDU **100**. For example, the indicators **210** are configured to indicate measured electrical characteristics of the PDU **100**, the status of the PDU **100**, etc. The one or more user input modules **220** may be operably coupled to the controller **200** to, for example, turn the PDU **100** on or off. In some embodiments, the one or more user input modules **220** may include a combination of digital and analog input or output devices required to achieve a desired level of operation for the PDU, such as one or more knobs, one or more dials, one or more switches, one or more buttons, etc. In some embodiments, the one or more user input modules **220** may receive signals wirelessly from a device external to the PDU **100** (e.g., a user's mobile phone).

[0051] The controller **200** is configured to selectively control the switches **235** to enable power output from one or more of the outlets **115**. The controller **200** is connected to the current sensors **225**, **230** for sensing and monitoring an overall current or power draw (e.g., electrical load) of the PDU **100** (e.g., using current sensor **225**), as well as individual current or power draws (e.g., electrical loads) from the individual outlets **115** (e.g., using respective current sensors **230**). If the overall current or power draw of the PDU **100** exceeds the user set current or power threshold for the PDU **100**, the controller is configured to disable or de-energize one or more of the outlets **115**. The controller **200** is also connected to the switches **235** for controlling an ON/OFF state of each of the switches **235**. Various processes for controlling the energization of the outlets **115** are described below.

[0052] FIG. 3 illustrates an example process 300 for controlling operation of the PDU 100. In the example process 300, the PDU 100 may be powered on (at block 302). For example, a user may power on the PDU 100 by selecting the ON/OFF button 125 and define a user set current threshold or a user set power threshold (e.g., via the user input 135). In the example process 300, the controller 200 determines user set current threshold or the user set power threshold (at block 304). In various implementations, the user set current threshold or the user set power threshold must be set every time the PDU 100 is turned ON. In some examples, the controller 200 initializes the user set current threshold or the user set power threshold to a default value when the PDU 100 is turned ON. In various implementations, the controller 200 stores one or more previous values for the user set current threshold or the user set power threshold and automatically sets the threshold to a stored value. The user may update the user set current threshold or the user set power threshold at any time (e.g., using user inputs 135).

[0053] In the example process 300, the controller 200 polls each of the outlets 115 to determine, for example, how much current and/or power each outlet 115 draws (at block 306). The current and/or power draw determined during the polling process may indicate how much power a device connected to the respective outlet 115 will draw during operation (e.g., during a charging cycle). The controller 200 may monitor a respective current sensor 230 associated with each outlet 115 to determine the current and/or power draw. In various implementations, the controller 200 determines the power draw by multiplying the current value sensed at the respective current sensor 230 with the voltage input via the power input cord 110 (e.g., via the power input module 205). In some examples, the controller 200 determines an average current draw over time and/or an average power draw over time. In various implementations, the controller 200 polls the outlets by turning on each outlet 115 individually, determining an average current draw and/or an average power draw, and then turning off the outlet 115. In some examples, the controller 200 polls the outlets 115 by turning on banks of two or more outlets 115, determining the average current draw and/or average power draw of each outlet 115, and turning off the bank of outlets 115. Additional processes for polling outlets 115 are described in greater detail below.

[0054] In the example process 300, the controller 200 determines which outlets 115 can be turned on without causing the total current draw to exceed the user set current threshold or the total power draw to exceed the user set power threshold and turns on each of the outlets 115 that have been determined can be turned on (at block 308). In various implementations, the controller 200 turns on as many outlets 115 as possible according to priority order. In some examples, the controller 200 maximizes the number of outlets 115 that are turned on. In various implementations, the controller 200 maximizes a total current draw or power draw of the outlets 115. Additional processes for turning on outlets 115 are described in greater detail below.

[0055] In the example process 300, once the outlets 115 are on, the controller 200 monitors the total current draw or total power draw of the PDU 100 (e.g., by monitoring the current sensor 225, by monitoring the current sensors 230 associated with each outlet that is turned on and summing the monitored current values, by monitoring all the current sensors 230 and summing the monitored current values, etc.) and compares the total current draw or total power draw of the PDU 100 with the user set current threshold or the user set power threshold (at decision block 310). In response to the total current draw or the total power draw of the PDU 100 exceeding the user set current threshold or the user set power threshold ("YES" at decision block 310), the controller 200 turns off one or more outlets 115 (at block 312). In various implementations, the controller 200 turns off the lowest priority outlet 115 that is powered on. In some examples, the controller 200 turns off banks of one or more outlets 115 grouped according to reverse-priority order.

[0056] In the example process 300, the controller 200 determines whether an outlet of interest has been turned off (at decision block 314). In various implementations, the outlet of interest may correspond to the outlet 115 that, after it is turned off, causes the total current draw or the total power draw of the PDU 100 to fall below the user set current threshold or the user set power

threshold. In response to the controller **200** determining that the outlet of interest has not been turned off (“NO” at decision block **314**), the controller **200** turns off one or more additional outlets **115** at block **312**. In response to the controller **200** determining that the outlet of interest has been turned off (“YES” at decision block **314**), the controller **200** again determines which outlets **115** can be turned on without causing the total current draw to exceed the user set current threshold or the total power draw to exceed the user set power threshold and turns on each of the outlets **115** that have been determined can be turned on at block **308**.

[0057] In various implementations, the controller **200** is configured to turn off outlets **115** until it reaches (and turns off) the outlet of interest more quickly than a wall circuit breaker can trip (e.g., according to a dynamic trip time). Thus, the time required for the controller **200** to turn off outlets **115** until it reaches (and turns off) the outlet of interest may depend on the thermal trip curve of common circuit breakers found in commercial and residential settings.

[0058] In the example process **300**, the controller **200** monitors the total current or power draw of the PDU **100**, compares the total current or power draw of the PDU **100** with the user set current threshold or the user set power threshold, and, in response to the total current draw or the total power draw of the PDU **100** not exceeding the user set current threshold or the user set power threshold (“NO” at decision block **310**), determines whether additional outlets **115** can be turned on and turned on each of the additional outlets that have been determined can be turned on (at decision block **316**). In response to determining that additional outlets **115** can be turned on (“YES” at decision block **316**), the controller **200** may turn on additional outlets **115** according to any of the previous described processes and/or the processes described in greater detail below (at block **318**).

[0059] After the controller **200** turns on additional outlets **115** (if able) at block **318**, the controller **200** again monitors the total current or power draw of the PDU **100** and compares the total current or power draw of the PDU **100** with the user set current threshold or the user set power threshold (at decision block **310**). In response to determining that additional outlets **115** cannot be turned on (“NO” at decision block **316**), the controller **200** again monitors the total current or power draw of the PDU **100** and compares the total current or power draw of the PDU **100** with the user set current threshold or the user set power threshold (at decision block **310**).

[0060] FIG. **4** illustrates an example process **400** for polling outlets **115** of the PDU **100**. In the example process **400**, the controller **200** turns on outlets **115** in groups of one or more starting from the highest priority outlets **115** to the lowest priority outlets **115**, measures the current draw of each of the outlets **115**, stores the current draw of each outlet **115** (e.g., in the memory **245**), and turns off the group of outlets **115** before repeating the process with the next group of outlets (e.g., until all outlets have been polled). While the process **400** is described with reference to current measurements, the process **400** may be implemented using power measurements instead.

[0061] In various implementations, current or power thresholds are not enforced during the polling process **400**. The stored current values may then be used by the controller **200** to determine which outlets **115** can be energized or turned on. In some examples, during the polling process **400**, each outlet **115** is turned on for a time period (e.g., one second) before it is turned back off. If a current value for the outlet **115** cannot be determined within the time period, the outlet may be assumed to be low risk of causing the PDU **100** to trip the circuit breaker. A fast load for an outlet **115** may be a load that reaches a steady state within the time period (e.g., a light, a fan, etc.). A slow load for the outlet **115** may be an outlet that does not reach a steady state within the time period (e.g., a battery pack charger). A no load outlet **115** (e.g., an outlet at which a current value is not detected) may be treated like a slow load outlet **115**.

[0062] In some examples, the polling process **400** is only performed once (e.g., upon startup of the PDU **100**). The stored values from previous polling processes can be overwritten or not used in favor of more recent, current measurements. For example, when an outlet **115** is on, a value for the current draw can be monitored and stored in the memory **245**. Thus, the stored current draw of the outlet **115** determined during the polling process **400** can be replaced with a more up-to-date value.

The polling process **400** shown in FIG. **4** illustrates the example where the PDU **100** has 12 outlets **115-1-115-12**. However, in other implementations, the PDU **100** may have any number of outlets **115**, and the process **400** may be scaled accordingly.

[0063] In the example process **400**, the controller **200** turns on outlets **115-1** and **115-2** (e.g., the two highest priority outlets) at block **402**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-1** and **115-2** and stores the respective current draw values to the memory **245** (at block **404**). In the example process **400**, the controller **200** turns the outlets **115-1** and **115-2** off (at block **406**). In the example process **400**, the controller **200** turns on outlets **115-3** and **115-4** (e.g., the two next-highest priority outlets) at block **408**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-3** and **115-4** and stores the respective current draw values to the memory **245** (at block **410**). In the example process **400**, the controller **200** turns the outlets **115-3** and **115-4** off (at block **412**).

[0064] In the example process **400**, the controller **200** turns on outlets **115-5** and **115-6** (e.g., the two next-highest priority outlets) at block **414**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-5** and **115-6** and stores the respective current draw values to the memory **245** (at block **416**). In the example process **400**, the controller **200** turns the outlets **115-5** and **115-6** off (at block **418**). In the example process **400**, the controller **200** turns on outlets **115-7** and **115-8** (e.g., the two next-highest priority outlets) at block **420**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-7** and **115-8** and stores the respective current draw values to the memory **245** (at block **422**). In the example process **400**, the controller **200** turns the outlets **115-7** and **115-8** off (at block **424**).

[0065] In the example process **400**, the controller **200** turns on outlets **115-9** and **115-10** (e.g., the two next-highest priority outlets) at block **426**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-9** and **115-10** and stores the respective current draw values to the memory **245** (at block **428**). In the example process **400**, the controller **200** turns the outlets **115-9** and **115-10** off (at block **430**). In the example process **400**, the controller **200** turns on outlets **115-11** and **115-12** (e.g., the two next-highest priority outlets) at block **432**. In the example process **400**, the controller **200** measures the currents drawn by the outlets **115-11** and **115-12** and stores the respective current draw values to the memory **245** (at block **434**). In the example process **400**, the controller **200** turns the outlets **115-11** and **115-12** off (at block **436**).

[0066] FIG. **5** illustrates an example process **500** for turning on outlets **115** of the PDU **100**. In various implementations, the controller **200** adds the outlets **115** that can be turned on without causing the PDU **100** to exceed the user set current threshold to a virtual queue and turns on each queued outlet **115** at the same time. While the process **500** is described with reference to current measurements, the process **500** may be implemented using power measurements instead. For example, the controller **200** cycles through the outlets **115** in priority order (e.g., from the highest priority to the lowest priority). The controller **200** may select an outlet **115** and determine whether the selected outlet **115** is on. In response to determining that the selected outlet **115** is on, the controller **200** moves on and selects the next outlet.

[0067] In response to determining that the selected outlet **115** is not on, the controller **200** determines whether turning on the selected outlet **115** (and any queued higher-priority outlets **115**) will cause the total current draw of the PDU **100** to exceed the user set current threshold. For example, the controller **200** may compute the total current draw of the PDU **100** using stored current draws (e.g., determined during the polling process) from the memory **245**. In various implementations, the controller **200** maintains a running total of what the total current draw of the PDU **100** should be when all queued outlets **115** are turned on. In response to determining that turning on the selected outlet **115** (and any queued higher-priority outlets **115**) will not cause the total current draw of the PDU **100** to exceed the user set current threshold, the controller **200** adds the selected outlet **115** to the queue (of outlets to be turned on) and selects the next outlet **115**.

[0068] In response to the controller **200** determining that turning on the selected outlet **115** (and

any queued higher-priority outlets **115**) will cause the total current draw of the PDU **100** to exceed the user set current threshold, the controller **200** does not add the selected outlet to the queue and instead selects the next outlet **115**. After cycling through each outlet **115**, the controller **200** may simultaneously turn on all of the queued outlets **115**. Thus, in various implementations, the controller **200** only turns on the outlets **115** that will keep the total current at or below the user set current threshold.

[0069] The process **500** shown in FIG. 5 illustrates the example where the PDU **100** has 12 outlets **115-1-115-12**. However, in other implementations, the PDU **100** may have any number of outlets, and the process **500** may be scaled accordingly. In the example process **500**, the controller **200** selects the highest priority outlet **115-1** and determines whether the selected outlet **115-1** is currently powered on (at decision block **502**). In response to determining that the selected outlet **115-1** is powered on (“YES” at decision block **502**), the controller **200** selects the next outlet **115-2** (e.g., the outlet **115** having the next highest priority). In response to determining that the selected outlet **115-1** is not powered on (“NO” at decision block **502**), the controller **200** determines whether turning the selected outlet **115-1** on will cause the total current draw of the PDU **100** to exceed the user set current threshold (at decision block **504**). In various implementations, the controller **200** compares the sum of the stored current value for the selected outlet **115-1** (e.g., from the memory **245**) and the total current draw of the PDU **100** with the user set current threshold.

[0070] In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-1** and the total current draw PDU **100** will exceed the user set threshold (“YES” at decision block **504**), the controller **200** selects the next outlet **115-2**. In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-1** and the total current draw PDU **100** will not exceed the user set threshold (“NO” at decision block **504**), the controller **200** adds the selected outlet **115-1** to the queue (at block **506**) and selects the next outlet **115-2**.

[0071] After the controller **200** selects the next outlet **115-2** (e.g., the outlet **115** having the next highest priority), the controller **200** determines whether the selected outlet **115-2** is currently powered on (at decision block **508**). In response to determining that the selected outlet **115-2** is powered on (“YES” at decision block **508**), the controller **200** selects the next outlet **115-3** (e.g., the outlet **115** having the next highest priority). In response to determining that the selected outlet **115-2** is not powered on (“NO” at decision block **508**), the controller **200** determines whether turning the selected outlet **115-2** on will cause the total current draw of the PDU **100** to exceed the user set current threshold (at decision block **510**). In various implementations, the controller **200** compares the sum of the stored current value for the selected outlet **115-2** (e.g., from the memory **245**), any outlets **115** in the queue, and the total current draw of the PDU **100** with the user set current threshold.

[0072] In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-2** and the total current draw PDU **100** will exceed the user set threshold (“YES” at decision block **510**), the controller **200** selects the next outlet **115-3**. In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-2** and the total current draw PDU **100** will not exceed the user set threshold (“NO” at decision block **510**), the controller **200** adds the selected outlet **115-2** to the queue (at block **512**) and selects the next outlet **115-3**. The controller **200** may proceed through the remaining outlets **115** of the PDU **100** (e.g., in priority order) as previously described. Accordingly, the description of the process **500** as applied to outlets **115-3-115-10** is omitted for the sake of conciseness.

[0073] After the controller **200** processes the outlet **115-10** and selects the next outlet **115-11** (e.g., the outlet **115** having the next highest priority), the controller **200** determines whether the selected outlet **115-11** is currently powered on (at decision block **514**). In response to determining that the selected outlet **115-11** is powered on (“YES” at decision block **514**), the controller **200** selects the next outlet **115-12** (e.g., the outlet **115** having the next highest priority). In response to determining

that the selected outlet **115-11** is not powered on (“NO” at decision block **514**), the controller **200** determines whether turning the selected outlet **115-11** on will cause the total current draw of the PDU **100** to exceed the user set current threshold (at decision block **516**). In various implementations, the controller **200** compares the sum of the stored current value for the selected outlet **115-11** (e.g., from the memory **245**), any outlets **115** in the queue, and the total current draw of the PDU **100** with the user set current threshold.

[0074] In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-11** and the total current draw PDU **100** will exceed the user set threshold (“YES” at decision block **516**), the controller **200** selects the next outlet **115-12**. In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-11** and the total current draw PDU **100** will not exceed the user set threshold (“NO” at decision block **516**), the controller **200** adds the selected outlet **115-11** to the queue (at block **518**) and selects the next outlet **115-12**.

[0075] After the controller **200** selects the next outlet **115-12** (e.g., the outlet **115** having the next highest priority), the controller **200** determines whether the selected outlet **115-12** is currently powered on (at decision block **520**). In response to determining that the selected outlet **115-12** is powered on (“YES” at decision block **520**), the controller **200** turns on the queued outlets **115** (at block **526**). In various implementations, the controller **200** turns on the queued outlets **115** simultaneously. In some examples, the controller **200** turns on the queued outlets in banks of one or more (for example, according to priority order or reverse priority order).

[0076] In response to determining that the selected outlet **115-12** is not powered on (“NO” at decision block **520**), the controller **200** determines whether turning the selected outlet **115-12** on will cause the total current draw of the PDU **100** to exceed the user set current threshold (at decision block **522**). In various implementations, the controller **200** compares the sum of the stored current value for the selected outlet **115-12** (e.g., from the memory **245**), any outlets **115** in the queue, and the total current draw of the PDU **100** with the user set current threshold.

[0077] In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-12** and the total current draw PDU **100** will exceed the user set threshold (“YES” at decision block **522**), the controller **200** turns on the queued outlets **115** (at block **526**). In response to the controller **200** determining that the sum of the stored current value for the selected outlet **115-12** and the total current draw PDU **100** will not exceed the user set threshold (“NO” at decision block **522**), the controller **200** adds the selected outlet **115-12** to the queue (at block **524**) and turns on the queued outlets **115** (at block **526**).

[0078] FIG. **6** illustrates an example process **600** for turning on outlets **115** of the PDU **100**. The process **600** is similar to the process **500**. However, instead of first queuing outlets **115** for simultaneous turn on (or turn on in banks), outlets **115** are sequentially turned on upon checking each outlet and determining that the outlet **115** can be turned on without exceeding the current threshold. While the process **600** is described with reference to current measurements, the process **600** may be implemented using power measurements instead. Thus, in the example process **600** of FIG. **6**, blocks **602-606** are similar to blocks **502-506** of the example process **500** of FIG. **5** except that at block **606**, the controller **200** turns the outlet **115-1** on instead of adding the outlet **115-1** to the queue.

[0079] In the example process **600** of FIG. **6**, blocks **608-612** are similar to blocks **508-512** of the example process **500** of FIG. **5** except that at block **612**, the controller **200** turns the outlet **115-2** on instead of adding the outlet **115-2** to the queue. In the example process **600** of FIG. **6**, blocks **614-618** are similar to blocks **514-518** of the example process **500** of FIG. **5** except that at block **618**, the controller **200** turns the outlet **115-11** on instead of adding the outlet **115-11** to the queue. In the example process **600** of FIG. **6**, blocks **620-624** are similar to blocks **520-524** of the example process **500** of FIG. **5** except that at block **624**, the controller **200** turns the outlet **115-12** on instead of adding the outlet **115-12** to the queue.

[0080] FIG. 7 illustrates an example process **700** for turning on outlets **115** of the PDU **100**. The process **700** is similar to the process **600**. However, in some implementations, the PDU **100** includes an additionally fully functional outlet, such as a thirteenth outlet **115-13**. The additional fully functional outlet **115-13** may be similar to the other outlets **115** (e.g., includes an associated current sensor **230**, switch **235**, etc.) but be generally inaccessible to the user during normal operation. In various implementations, the additional fully functional outlet **115-13** may be positioned within the housing **105**, covered by a covering, etc. In some examples, the additional fully functional outlet **115-13** may be used for future expansion of the internal components of the PDU **100**, used during manufacturer testing, etc. In various implementations, the additional fully functional outlet **115-13** is the lowest priority outlet **115**.

[0081] While the process **700** is described with reference to current measurements, the process **700** may be implemented using power measurements instead. In the example process **700** of FIG. 7, blocks **702-712** are similar to blocks **602-612** of the example process **600** of FIG. 6, and blocks **714-718** are similar to blocks **620-624**. However, in the example process **700**, the controller **200** selects the additional fully functional outlet **115-13** after the controller **200** processes the outlet **115-12**. For example, the controller **200** selects the additional fully-functional outlet **115-13** in response to determining that the outlet **115-12** is on (“YES” at decision block **714**), in response to determining that turning on the outlet **115-12** would cause the total current draw of the PDU **100** to exceed the user set current threshold (“YES” at decision block **716**), or after turning on the outlet **115-12** at block **718**.

[0082] After selecting the fully functional outlet **115-13**, the controller **200** determines whether the selected additional fully functional outlet **115-13** is currently powered on (at decision block **720**). In response to determining that the selected additional fully functional outlet **115-13** is powered on (“YES” at decision block **720**), the process **700** returns to the main process **300**. In response to determining that the selected additional fully functional outlet **115-13** is not powered on (“NO” at decision block **720**), the controller **200** determines whether turning the additional fully functional outlet **115-13** on would cause the total current draw of the PDU **100** to exceed the user set current threshold (at decision block **722**). In response to determining that turning the additional fully functional outlet **115-13** on would cause the total current draw of the PDU **100** to exceed the user set current threshold (“YES” at decision block **722**), the process **700** returns to the main process **300**.

[0083] In response to determining that turning the additional fully-functional outlet **115-13** on would not cause the total power draw of the PDU **100** to exceed the user set current threshold (“NO” at decision block **722**), the controller **200** determines whether all of the higher priority outlets **115** of the PDU **100** are on (at decision block **724**). In various implementations, the higher priority outlets **115** may include all of the outlets **115** of the PDU **100** aside from the additional fully-functional outlet **115-13**, such as outlets **115-1-115-12**. In response to determining that all of the higher priority outlets **115** are on (“YES” at decision block **724**), the controller **200** does not turn the additional fully functional outlet **115-13** on and the process **700** returns to the main process **300**. In response to determining that all of the higher priority outlets **115** are not on (“NO” at decision block **724**), the controller **200** turns the additional fully functional outlet **115-13** on at block **726** and the process **700** returns to the main process **300**.

[0084] FIG. 8 illustrates an example process **800** for turning on outlets **115** of the PDU **100**. The process **800** of FIG. 8 is similar to the process **500** of FIG. 5. However, in the process **800**, after the controller **200** checks a pair of outlets **115** (e.g., outlets **115-1** and **115-2**, outlets **115-3** and **115-4**, etc.), the controller **200** will turn on both outlets **115** or none of the two outlets **115**. For example, if either outlet **115** of a pair of outlets **115** is queued, then both outlets **115** of the pair will be turned on. If neither outlet **115** of the pair is queued, then neither outlet **115** of the pair will be turned on. This process **800** may be repeated for each of the outlet pairs in the PDU **100** (e.g., for each of the six outlet pairs in the twelve outlet PDU **100** illustrated in FIG. 1). While the process **800** is

described with reference to current measurements, the process **800** may be implemented using power measurements instead.

[0085] Thus, in the example process **800**, blocks **802-812** are similar to blocks **502-512** of the example process **500**. However, in the example process **800**, after the controller **200** processes the first pair of outlets **115-1** and **115-2** at blocks **802-812**, the controller **200** turns both outlets **115-1** and **115-2** on at block **814** in response to either of the outlets **115-1** or **115-2** being added to the queue at block **806** or **812**. In the example process **800**, blocks **816-826** are similar to blocks **514-524** of the example process **500**. However, in the example process **800**, after the controller **200** processes the first pair of outlets **115-11** and **115-12** at blocks **802-812**, the controller **200** turns both outlets **115-11** and **115-12** on at block **828** in response to either of the outlets **115-11** or **115-12** being added to the queue at block **820** or **826**.

[0086] FIG. **9** illustrates a process **900** for controlling operation of the PDU **100**. The process **900** of FIG. **9** is similar to the process **300** of FIG. **3**. For example, blocks **902-918** of the process **900** may be similar to blocks **302-318** of the process **300**. However, the process **900** further includes a timer. For example, the process **900** includes a decision block **920** (implementing a timer) between blocks **916** and **918**. The timer may be set, for example, each time an outlet **115** is turned on or each time a pair of outlets **115** (e.g., outlets **115-1** and outlets **115-2**) are queued for turn on. The timer functionality introduces a delay that prevents outlets **115** from being turned on too fast. For example, after the controller **200** determines that additional outlets can be turned on (“YES” at decision block **916**), the controller **200** determines whether the timer has finished (at decision block **920**). The controller **200** may turn on the additional outlets **115** at block **918** only in response to the timer being finished (“YES” at decision block **916**).

[0087] FIG. **10** illustrates a process **1000** for turning on outlets **115** of the PDU **100**. The process **1000** may be a continuation of the process **900** (for example, describing blocks **908** and/or **918** in additional detail). The process **1000** of FIG. **10** may be similar to the process **800** of FIG. **8**. However, in the process **1000**, the timer is set in response to a pair of outlets **115** being turned on. For example, blocks **1002-1012** of the process **1000** may be similar to blocks **802-812** of the process **800**. However, in the process **1000**, after processing the first outlet pair (e.g., outlets **115-1** and **115-2**) at blocks **1002-1012**, the controller **200** determines whether either outlet **115-1** or **115-2** were added to the queue at decision block **1014**. In response to determining that neither outlet **115-1** nor **115-2** were added to the queue (“NO” at decision block **1014**), the controller **200** continues processing the next pair of outlets **115** (e.g., outlets **115-3** and **115-4**). In response to determining that either outlet **115-1** or **115-2** were added to the queue (“YES” at decision block **1014**), the controller **200** turns on the outlets **115-1** and **115-2** (at block **1016**) and sets the timer (at block **1018**) before proceeding to process the next pair of outlets **115**.

[0088] Blocks **1020-1030** of the process **1000** may be similar to blocks **816-826** of the process **800**. However, in the process **1000**, after processing the final outlet pair (e.g., outlets **115-11** and **115-12**) at blocks **1020-1030**, the controller **200** determines whether either outlet **115-11** or **115-12** were added to the queue at decision block **1032**. In response to determining that neither outlet **115-11** nor **115-12** was added to the queue (“NO” at decision block **1032**), the process **1000** returns to the main process **900**. In response to determining that either outlet **115-11** or **115-12** were added to the queue (“YES” at decision block **1032**), the controller **200** turns on the outlets **115-11** and **115-12** (at block **1034**) and sets the timer (or resets the timer if the timer is already running) at block **1036** before the process **1000** proceeds back to the main process **900**.

[0089] FIGS. **11-13** illustrate a process **1100** for controlling operation of the PDU **100**. In the example process **1100**, the PDU **100** may be powered on (at block **1102**). In the example process **1100**, the controller **200** determines the user set current threshold (at block **1104**). While the process **1100** is described with reference to current measurements, the process **1100** may be implemented using power measurements instead. In the example process **1100**, the controller **200** monitors the total current draw of the PDU **100** and determines whether the total current draw

exceeds a critical threshold (at decision block **1106**). In various implementations, the critical threshold is a pre-programmed threshold and may be set to a value greater than or equal to the maximum programmable user set current threshold. In some examples, the critical threshold may be set to a value above the rated current limit of the circuit that the PDU **100** is plugged in to (e.g., **16A** for a **15A** circuit).

[0090] In response to determining that the total current draw of the PDU **100** exceeds the critical threshold (“YES” at decision block **1106**), the controller **200** turns off high load outlets **115** (at block **118**). High load outlets **115** may be outlets drawing current exceeding a threshold value. In various implementations, the threshold value is about 4 A. In some examples, the controller **200** turns off all outlets **115** drawing more current than the threshold value simultaneously. In various implementations, the controller **200** turns off outlets **115** drawing more current than the threshold in blocks of one or more outlets (e.g., in reverse priority order). After the controller **200** turns off high load outlets at block **1108**, the process **1100** returns to block **1104**.

[0091] In response to determining that the total current draw of the PDU **100** does not exceed the critical threshold (“NO” at decision block **1106**), the controller **200** determines whether the total current draw of the PDU **100** exceeds the user set current threshold (at decision block **1110**). In response to the total current draw of the PDU **100** exceeding the user set current threshold (“YES” at decision block **1110**), the controller **200** selects the lowest priority outlet **115** at block **1112** (see FIG. **12**) and determines whether the selected outlet **115** is energized (at decision block **1114**). In response to determining that the selected outlet **115** is energized (“YES” at decision block **1114**), the controller **200** turns off the selected outlet **115** at block **1116** and the process **1100** returns to block **1104**.

[0092] In response to determining that the selected outlet **115** is not energized (“NO” at decision block **1114**), the controller **200** determines whether the highest priority outlet **115** is selected (at decision block **1118**). In response to determining that the highest priority outlet **115** is selected (“YES” at decision block **1118**), the process **1100** returns to block **1104**. In response to determining that the highest priority outlet **115** is not selected (“NO” at decision block **1118**), the controller **200** selects the outlet **115** having the next highest priority at block **1120** and the process **1100** returns to decision block **1114**.

[0093] In response to the total current draw of the PDU **100** not exceeding the user set current threshold (“NO” at decision block **1110**), the controller **200** selects the highest priority outlet **115** (at block **1122**) (see FIG. **13**). In the example process **1100**, the controller **200** determines whether the selected outlet **115** is energized (at decision block **1124**). In response to determining that the selected outlet is not energized (“NO” at decision block **1124**), the controller **200** determines whether the lowest priority outlet **115** is selected (at decision block **1126**). In response to determining that the lowest priority outlet **115** is selected (“YES” at decision block **1126**), the process **1100** returns to block **1104**. In response to determining that the lowest priority outlet **115** is not selected (“NO” at decision block **1126**), the controller **200** selects the next outlet **115** at block **1128** and the process returns to decision block **1124**.

[0094] In response to determining that the selected outlet **115** is energized (“YES” at decision block **1124**), the controller **200** determines whether the selected outlet **115** was previously tripped (e.g., previously turned off in response to the overall current draw of the PDU **100** exceeding the user set current threshold) at decision block **1130**. In response to determining that the selected outlet was not previously tripped (“NO” at decision block **1130**), the controller **200** turns on the selected outlet **115** at block **1132** and the process returns to block **1104**. In response to determining that the selected outlet was previously tripped (“YES” at decision block **1130**), the controller **200** determines whether the total current draw of the PDU **100** plus an additional safety margin exceeds the user set threshold current (at decision block **1134**). In various implementations, the addition safety margin may be a sum of the current draw of the selected outlet **115** when it was previously tripped plus an additional value. In some examples, the additional value may be about

1.5 A.

[0095] In response to determining that the total current draw of the PDU **100** plus the additional safety margin exceeds the user set threshold current (“YES” at decision block **1134**), the process **1100** returns to decision block **1126**. In response to determining that the total current draw of the PDU **100** plus the additional safety margin does not exceeds the user set threshold current (“NO” at decision block **1134**), the controller **200** turns on the selected outlet **115** at block **1136** and the process **1100** returns to block **1104**.

[0096] FIG. **14** illustrates a process **1400** for controlling operation of the PDU **100**. In the example process **1100**, the PDU **100** may be powered on (at block **1402**). In the example process **1400**, the controller **200** determines the user set current threshold (at block **1404**). While the process **1400** is described with reference to current measurements, the process **1400** may be implemented using power measurements instead. In the example process **1400**, the controller **200** polls the outlets **115** (at block **1406**). For example, the controller **200** polls the outlets according to any of the previously described techniques. In various implementations, the controller **200** turns on each outlet **115** for a period of time (e.g., about 1 second), measures the current value of the outlet (e.g., an average value over the period of time), turns off the outlet **115**, and selects the next outlet **115** for polling. In some examples, the controller **200** polls the outlets **115** according to priority order. In various implementations, the controller **200** polls the outlets **115** according to reverse priority order.

[0097] In the example process **1400**, the controller **200** turns on selected outlets **115** (at block **1408**). In various implementations, the controller **200** first selects a combination of outlets **115** that can be turned on without causing the overall current draw of the PDU **100** to exceed the user set current threshold, then turns on the selected outlets **115** simultaneously. In some examples, the controller **200** selects a combination of outlets **115** according to priority (e.g., higher priority outlets **115** are selected before lower priority outlets **115**). In various implementations, the controller **200** selects a combination of outlets **115** to maximize a number of outlets **115** selected. In some examples, the controller **200** selects a combination of outlets **115** to maximize the total current draw of the PDU **100**. In various implementations, the controller **200** selects a combination of outlets based on a combination of the previously described factors.

[0098] In the example process **1400**, the controller **200** determines whether the total current draw of the PDU **100** exceeds the user set current threshold (at decision block **1410**). In response to the total current draw of the PDU **100** exceeding the user set current threshold (“YES” at decision block **1410**), the controller **200** turns off all outlets **115** of the PDU **100** at block **1412**, and the controller **200** determines whether a re-polling condition is met (at decision block **1414**). In various implementations, the re-polling condition is met after a pre-determined time duration has elapsed since the outlets **115** were last polled. In some examples, the re-polling condition is met after the outlets **115** are turned off in response to the total current draw of the PDU **100** exceeding the user set current threshold (e.g., a trip condition). In various implementations, the re-polling condition is met in response to the controller **200** determining that one or more outlets **115** have erratic current readings.

[0099] In response to the re-polling condition being met (“YES” at decision block **1414**), the process **1400** returns to block **1404**. In response to the re-polling condition not being met (“NO” at decision block **1414**), the process **1400** returns to block **1408**. In response to the total current draw of the PDU **100** not exceeding the user set current threshold (“NO” at decision block **1410**), the controller **200** determines whether any additional outlets **115** can be turned on (at decision block **1416**). In various implementations, the controller **200** determines whether any additional outlets **115** can be turned on based on any of the previously described techniques (e.g., with reference to block **1408**). In response to the controller **200** determining that one or more additional outlets **115** can be turned on (“YES” at decision block **1416**), the controller **200** turns on the additional outlets at block **1418** and the process **1400** returns to decision block **1414**. In response to the controller **200** determining that no additional outlets **115** can be turned on (“NO” at decision block **1416**), the

process **1400** returns to decision block **1414**.

[0100] Thus, embodiments described herein provide, among other things, a power distribution unit including a plurality of intelligently controlled outlets. Various features and advantages are set forth in the following claims.

Claims

1. A power distribution unit comprising: a plurality of outlets; and a controller configured to: poll the plurality of outlets to determine a polled electrical load of each outlet, add one or more outlets of the plurality of outlets to a queue based on a comparison of the polled electrical loads and a user set threshold, connect the queued outlets to a power input, monitor a total electrical load of the power distribution unit, and disconnect, in response to determining that the total electrical load exceeds the user set threshold, one or more outlets of the plurality of outlets from the power input.
2. The power distribution unit of claim 1, wherein, to poll the plurality of outlets, the controller is configured to: connect a first pair of outlets of the plurality of outlets to the power input; measure an electrical load of each outlet of the first pair of outlets; and disconnect the first pair of outlets of the plurality of outlets from the power input.
3. The power distribution unit of claim 2, wherein: each outlet of the plurality of outlets is assigned a priority; and to poll the plurality of outlets, the controller is configured to: connect a second pair of outlets of the plurality of outlets to the power input, the second pair of outlets having a lower priority than the first pair of outlets, measure an electrical load of each outlet of the second pair of outlets, and disconnect the second pair of outlets of the plurality of outlets from the power input.
4. The power distribution unit of claim 1, wherein, to add one or more outlets of the plurality of outlets to the queue, the controller is configured to: select a first outlet of the plurality of outlets; determine whether connecting the first outlet to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold; and add, in response to determining that connecting the first outlet to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold, the first outlet to the queue.
5. The power distribution unit of claim 4, wherein: each outlet of the plurality of outlets is assigned a priority; and to add one or more outlets of the plurality of outlets to the queue, the controller is configured to: select a second outlet of the plurality of outlets, the second outlet having a lower priority than the first outlet, determine whether connecting the second outlet and any queued outlets to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold; and add, in response to determining that connecting the second outlet and any queued outlets to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold, the second outlet to the queue.
6. The power distribution unit of claim 1, wherein the controller is configured to disconnect outlets from the power input in reverse priority order in response to determining that the total electrical load exceeds the user set threshold.
7. The power distribution unit of claim 1, wherein the controller is further configured to: determine, in response to determining that the total electrical load does not exceed the user set threshold, whether any additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold; initiate, in response to determining that additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold, a timer; and connect, in response to the timer elapsing, the additional outlets to the power input.
8. The power distribution unit of claim 1, further comprising: a plurality of switches, each switch of the plurality of switches being associated with a respective outlet of the plurality of outlets; a plurality of outlet sensors, each outlet sensor of the plurality of sensors being associated with a respective outlet of the plurality of outlets and configured to sense an electrical load of the

respective outlet; and a main sensor configured to sense the total electrical load of the power distribution unit at the power input; wherein the controller is configured to: actuate each switch of the plurality of switches to connect or disconnect the respective outlet, determine an electrical load of each outlet by monitoring a respective outlet sensor, and determine the total electrical load of the power distribution unit by monitoring the main sensor.

9. The power distribution unit of claim 1, wherein the polled electrical loads and the total electrical load are current values.

10. The power distribution unit of claim 1, wherein the polled electrical loads and the total electrical load are power values.

11. A method for controlling a power distribution unit, the method comprising: polling a plurality of outlets of the power distribution unit to determine a polled electrical load of each outlet; adding one or more outlets of the plurality of outlets to a queue based on a comparison of the polled electrical loads and a user set threshold; connecting the queued outlets to a power input; monitoring a total electrical load of the power distribution unit; and disconnecting, in response to determining that the total electrical load exceeds the user set threshold, one or more outlets of the plurality of outlets from the power input.

12. The method of claim 11, wherein polling the plurality of outlets includes: connecting a first pair of outlets of the plurality of outlets to the power input; measuring an electrical load of each outlet of the first pair of outlets; and disconnecting the first pair of outlets of the plurality of outlets from the power input.

13. The method of claim 12, wherein: each outlet of the plurality of outlets is assigned a priority; and polling the plurality of outlets includes: connecting a second pair of outlets of the plurality of outlets to the power input, the second pair of outlets having a lower priority than the first pair of outlets, measuring an electrical load of each outlet of the second pair of outlets, and disconnecting the second pair of outlets of the plurality of outlets from the power input.

14. The method of claim 11, wherein adding one or more outlets of the plurality of outlets to the queue includes: selecting a first outlet of the plurality of outlets; determining whether connecting the first outlet to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold; and adding, in response to determining that connecting the first outlet to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold, the first outlet to the queue.

15. The method of claim 14, wherein: each outlet of the plurality of outlets is assigned a priority; and adding one or more outlets of the plurality of outlets to the queue includes: selecting a second outlet of the plurality of outlets, the second outlet having a lower priority than the first outlet, determining whether connecting the second outlet and any queued outlets to the power input will cause the total electrical load of the power distribution unit to exceed the user set threshold; and adding, in response to determining that connecting the second outlet and any queued outlets to the power input will not cause the total electrical load of the power distribution unit to exceed the user set threshold, the second outlet to the queue.

16. The method of claim 11, further comprising: disconnecting outlets from the power input in reverse priority order in response to determining that the total electrical load exceeds the user set threshold.

17. The method of claim 11, further comprising: determining, in response to determining that the total electrical load does not exceed the user set threshold, whether any additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold; initiating, in response to determining that additional outlets of the plurality of outlets can be connected to the power input without causing the total electrical load to exceed the user set threshold, a timer; and connecting, in response to the timer elapsing, the additional outlets to the power input.

18. The method of claim 11, wherein the power distribution unit includes: a plurality of switches,

each switch of the plurality of switches being associated with a respective outlet of the plurality of outlets; a plurality of outlet sensors, each outlet sensor of the plurality of sensors being associated with a respective outlet of the plurality of outlets and configured to sense an electrical load of the respective outlet; a main sensor configured to sense the total electrical load of the power distribution unit at the power input; and the method further comprising: actuating each switch of the plurality of switches to connect or disconnect the respective outlet, determining an electrical load of each outlet by monitoring a respective outlet sensor, and determining the total electrical load of the power distribution unit by monitoring the main sensor.

19. The method of claim 11, wherein the polled electrical loads and the total electrical load are current values.

20. The method of claim 11, wherein the polled electrical loads and the total electrical load are power values.
